

A PROJECT TITLE

ON

Early-Stage Diabetes Risk Prediction Using Random Forest and XGBoost Classifiers
project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

Name Kolavali Sanjana roll 2520030614

Name Padala Anikha roll 2520030613

Under the Esteemed guidance of

Jyothsna Datti

Assistant Professor

CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Early-Stage Diabetes Risk Prediction Using Random Forest and XGBoost Classifiers

1. Abstract:

Diabetes is a common health problem, and identifying its risk at an early stage can help people take necessary precautions and seek medical advice on time. This project focuses on developing a machine learning-based system for early-stage diabetes risk prediction using Random Forest and XGBoost classifiers. The Early Stage Diabetes Risk Prediction Dataset from the UCI Machine Learning Repository will be used for this project. The dataset contains 520 patient records with 16 input features, including age, gender, polyuria, polydipsia, sudden weight loss, weakness, obesity, and other diabetes-related symptoms. The data will be preprocessed by handling categorical values, checking the data, and preparing the features for model training. Both Random Forest and XGBoost models will be trained using the prepared dataset and their results will be compared. The performance of the models will be evaluated using accuracy, precision, recall, F1-score, and confusion matrix. The comparison will help identify which model gives better results for predicting diabetes risk. The main aim of this project is to develop a simple and effective prediction system that can identify possible diabetes risk based on commonly available patient information. The system is intended to work as an early screening tool and not as a replacement for professional medical diagnosis. Through this project, machine learning techniques are applied to healthcare data to understand how patient symptoms and other factors can be used for early diabetes risk prediction.

2. INTRODUCTION:

Diabetes is a major health concern that can affect a person's daily life and may lead to other health complications if it is not identified and managed properly. One of the main reasons for choosing this project is the importance of identifying diabetes risk at an early stage. Many people may not immediately recognize the symptoms or may ignore them until the condition becomes more serious. With the increasing availability of healthcare data, machine learning can be used to study different patient factors and find patterns that may help in early risk prediction.

Machine learning is widely used for classification and prediction problems because it can learn patterns from existing data and use them to make predictions on new data. In this project, Random Forest and XGBoost are selected to predict whether a person is at risk of diabetes. The Early Stage Diabetes Risk Prediction Dataset from the UCI Machine Learning Repository contains 520 patient records and information related to age, gender, and different symptoms such as polyuria, polydipsia, weakness, sudden weight loss, itching, obesity, and visual blurring.

The dataset will be cleaned and converted into a suitable format before training the models. Random Forest will be used to build multiple decision trees and combine their results, while XGBoost will use a boosting approach to improve the classification results. The two models will be compared using accuracy, precision, recall, F1-score, and confusion matrix. The final objective is to identify the better-performing model and develop a system that can provide an early indication of diabetes risk based on the given patient information.

3. SOFTWARE REQUIREMENTS:

The project requires a basic computer system capable of running Python and the required machine learning libraries. Python will be used as the main programming language for data preprocessing, model development, and evaluation. Jupyter Notebook or Google Colab can be used as the development environment. Libraries such as Pandas and NumPy will be used for handling and processing the dataset, while Matplotlib and Seaborn will be used for data

visualization. Scikit-learn will be used for preprocessing, Random Forest, and model evaluation, and XGBoost will be used to implement the XGBoost classifier. A standard computer with sufficient RAM and storage is enough for this project since the selected dataset is relatively small.

CATEGORY	REQUIREMENT
Operating System	Windows 10 / Windows 11
Programming Language	Python 3.x
Development Environment	Jupyter Notebook / Google COLab
Libraries	Padas, Numpy, Matplotlib, Seaborn, Scikit-learn, XGBoost
Processor	Intel Core i3 or above
RAM	Minimum 4 GB
Storage	Minimum 2 GB free space
Dataset	UCI Early Stage Diabetes Risk Prediction Dataset
Machine Learning Models	Random Forest, XGBoost
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, Confusion Matrix

Signature of the Guide

Course Instructor